MEA CO2 Capture Ver 1 · operator training simulator

PROCUREMENT TO COMMISSIONING PLAN

Packages, FAT/SAT, completion, pre-commissioning, commissioning, performance test, handover

MEA CO2 capture unit: absorber T-101 / stripper T-102 (30 wt% MEA)

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Teaching document generated for the MEA CO2 capture operator training simulator. Not for construction. Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

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Body pages are A4 landscape. The same data file (cm.json) feeds the Procurement to commissioning tab of the OTS, where each commissioning step can be practised on the simulator.

1 · Basis and sequence

Numbers come from the MEA CO2 Capture Ver 1 simulator model (design point, noise off); items marked TBC/assumed are not design data.

Sequence: procurement (purchase orders, vendor drawings) → FAT at the vendor → construction and pressure testing → mechanical completion per system (punch A closed) → pre-commissioning (flushing, passivation, leak test, loop checks, SIS proof test, N_2 purge, amine make-up) → ready for start-up (RFSU) → commissioning → performance test → handover.

2 · Procurement packages and long-lead items

Package	Scope	Tags	Specified from	Lead, weeks (assumed)	Inspection
PKG-01 Columns and internals	Absorber T-101 (D 5.4 m, 2 × 8.5 m Mellapak Plus 252Y + 3 m water wash), stripper T-102 (D 2.6 m, 8 m IMTP 50), reflux drum V-101; liquid distributors, collectors, support plates, demister	T-101, T-102, V-101	Design tab (udesign), P&ID; notes 6 and 9; design pressures and materials TBC (Mechanical tab)	40–52 **long lead**: Large-diameter shell fabrication and structured packing in SS 316L	Shop: material certificates, weld NDT, hydrotest, dimensional check; packing and distributor inspection
PKG-02 Heat exchangers	Lean/rich plate exchanger E-101 (18.4 MW, SS 316L), kettle reboiler E-102 (13.7 MW, LP steam), condenser E-103, lean cooler E-104, wash cooler E-105	E-101, E-102, E-103, E-104, E-105	Process tab heat balance; model UA values (K.UAlr, K.UAreb, K.UAc, K.UAlc); P&ID; equipment list	30–44 **long lead**: E-101 plate pack and E-102 kettle shell are the longest-lead exchangers	Shop: hydrotest / plate-pack pressure test, material certificates, thermal rating review
PKG-03 Rotating equipment	Flue gas blower K-101 with 355 kW motor and VFD; centrifugal pumps P-101 to P-104 with motors (spares TBC)	K-101, P-101, P-102, P-103, P-104	Electrical load list MEA-EE-LL-001 (powers, motor ratings); model blower curve and flows	36–52 **long lead**: K-101 blower, its motor and the MV VFD	Shop: mechanical run test, performance curve, vibration, NPSH (pumps); VFD combined test
PKG-04 Valves and instruments	Control valves (9), shut-off valve XV-207 and flue-gas damper XV-101, transmitters, flow elements, SIS trip switches, CO2 and solvent-loading analysers	FV-109, FV-201, FV-207, FV-208, FV-303, TV-110, TV-202, TV-302, PV-301, XV-101, XV-207, AIT-101, AIT-111, AIT-113, AIT-203, AIT-213	Instrumentation & control tab (instrument list), C&E; MEA-IN-CE-001, P&ID; fail positions (note F)	24–40 **long lead**: Large flue-gas damper XV-101 and the solvent-loading analysers AIT-113 / AIT-203	Shop: seat leakage, stroke and fail-position test, calibration certificates; analyser FAT with test gases / standards
PKG-05 SIS logic solver	SIL-capable logic solver for interlocks I-1, I-2, I-3, ESD pushbutton, first-out, bypass/maintenance-override management	I-1, I-2, I-3, HS-ESD	SRS MEA-IN-SRS-001, C&E; MEA-IN-CE-001 (SRS of SIF and C&E; tabs)	26–40 **long lead**: Application software is engineered and tested against the C&E; before shipping	FAT: every C&E; cause → effect with simulated I/O, delays, first-out, reset logic
PKG-06 DCS (basic process control system)	Controllers, operator graphics, alarm list, historian; the OTS screens are the template	11 loops, alarms	Control loop design tab, ALM list, P&ID;	26–36	FAT: graphics against the P&ID;, every loop in simulation, alarm settings against the alarm list
PKG-07 Electrical	Transformer TR-101 6.6/0.4 kV 250 kVA, MCC-101 400 A, K-101 MV feeder and VFD, cables, lighting, earthing, heat tracing	TR-101, MCC-101, VFD K-101, cables	Electrical load list MEA-EE-LL-001	20–36	Shop: transformer routine tests, MCC type-test certificates and functional test
PKG-08 Piping and bulk materials	Pipe, fittings, flanges and manual valves; SS 316L for hot or loaded amine, CS + CA for cold lean (TBC)	13 line numbers	P&ID; line numbers (sizes TBC), materials note 6	16–26	Material certificates (PMI on SS), dimensional
PKG-09 First fill chemicals and utilities	MEA for the first solvent inventory, demin water, nitrogen for purging and blanketing, antifoam	MEA, N2, antifoam	Inventory from the model (part 5)	4–8	Certificate of analysis (MEA purity, chlorides, iron)

3 · FAT and SAT plan

Package	Item	FAT (at the vendor)	SAT (on site)	Acceptance
PKG-01	Columns	Hydrotest of shells, NDT records, internals trial fit, distributor levelness	Distributor water test after installation (even flow), packing installed per drawings, nozzle orientation	No leaks; distributor flow deviation within the vendor limit
PKG-02	Exchangers	Hydrotest / plate-pack pressure test each side, material and weld records	Site leak test with the system, gasket check on E-101 after transport	No leaks; no cross-leak between lean and rich sides of E-101
PKG-03	K-101 blower + VFD	Mechanical run test at speed, vibration and bearing temperatures, VFD + motor combined test	Rotation check, uncoupled motor run, coupled run on air, vibration baseline, trip from I-1	Vibration within the vendor limit; I-1 stops K-101
PKG-03	Pumps P-101…P-104	Performance test (head-flow, efficiency, NPSH required), hydrotest of casing	Rotation, solo motor run, water run on recirculation, seal check	Duty point reached; no seal leaks
PKG-04	Control and shut-off valves	Seat leakage, stroke time, fail-position on loss of air	Stroke from the DCS / SIS, fail position on air loss, positioner calibration	Fail positions as P&ID; note F; XV close time within the SRS
PKG-04	Analysers AIT-101/111/113/203/213	Calibration with test gases / standard solutions, response time	Re-calibration on site, sample system leak check, comparison with lab samples	Within the stated accuracy against lab
PKG-05	SIS logic solver	Every C&E; row: cause simulated → effects, delay, first-out, latch and reset	Loop checks to field devices, then a full proof test of each SIF	100 % of C&E; rows pass; delays as specified
PKG-06	DCS	Graphics against the P&ID;, every loop in simulation, alarm list, cascade TIC-206 → FIC-207	Loop checks (each tag from the field to the screen), control valve strokes	Every loop and alarm tested and signed
PKG-07	Electrical	Transformer routine tests, MCC functional test	Insulation resistance, protection relay settings, energisation procedure	Test sheets signed before energisation

4 · Mechanical completion and punch lists

System	Name	Main items
SYS-01	Flue gas	K-101, duct DUCT-FG-101, XV-101, FIC-101, AIT-101
SYS-02	Absorber and water wash	T-101, P-104, E-105, TV-110, TIC-110, AIT-111
SYS-03	Rich amine	P-101, FV-109, E-101 cold side, LIC-109, LSHH/LSLL-109, AIT-113
SYS-04	Stripper and reboiler	T-102, E-102, FV-207, XV-207, TIC-206, PSHH-301, TSHH-206, LSLL-208
SYS-05	Lean amine	P-102, E-101 hot side, E-104, TV-202, FV-201, AIT-203, AIT-213
SYS-06	Overhead and CO2 product	E-103, V-101, P-103, TV-302, FV-303, PV-301, FI-304
SYS-07	Utilities	Cooling water, LP steam, demin make-up (FV-208), nitrogen, instrument air
SYS-08	Electrical, DCS and SIS	TR-101, MCC-101, VFD, DCS, SIS logic solver, cabling

Punch	Meaning	Examples on this unit
A	Must be cleared before pre-commissioning of the system (safety or function)	Missing PSV on T-102 (P&ID; note 5: HOLD); XV-207 fails open on air loss; LSLL-208 not wired; flange bolting not torqued
B	Can be cleared during commissioning or after start-up without risk	Missing insulation on a cold line; tag plate missing; one spare cable gland
C	Cosmetic or documentation	Paint touch-up; as-built mark-up not yet transferred

5 · Pre-commissioning checklist

No.	Activity	What is done	Systems	Acceptance
PC-01	Utilities commissioned	CW, LP steam, demin water, instrument air and nitrogen available at the battery limit; LP steam lines blown clean to E-102	SYS-07	Blow-out targets clean; utility pressures as the basis (LP steam 3.5 bar a TBC)
PC-02	Flushing	Water flush of amine, wash-water and reflux lines at ≥ 1.5 m/s (typical) until the outlet runs clean; strainers fitted on pump suctions	SYS-02…06	Clean target / strainer after the last flush
PC-03	Cleaning and passivation of SS 316L	Degrease, then passivate stainless lines and E-101 (nitric or citric acid, ASTM A967 typical); rinse with demin water	SYS-03…06	Chloride in the rinse below the specification (TBC)
PC-04	Leak test	System leak test with water or nitrogen after reinstatement of all joints (pressure below the PSV set; PSVs on HOLD, note 5)	all	No pressure decay over the hold time; every joint signed
PC-05	Loop checks	Every instrument from the field device to the DCS / SIS screen; every control valve stroked; fail positions on air loss	SYS-08	Each loop sheet signed (11 control loops, all indicators and analysers)
PC-06	SIS proof test	Each trip of the C&E; tested from the sensor to the final element with the real delay; first-out and reset checked	SYS-08	All SIFs pass; trip delays as the SRS (3–5 s)
PC-07	Motor and drive checks	Insulation test, rotation, solo run of every motor on the load list; K-101 VFD tuned	SYS-01…06	Rotation correct; currents and vibration recorded
PC-08	Water circulation run	Circulate demin water with P-101 / P-102 through both columns, heat with steam, test the level and flow loops on water	SYS-02…06	Loops stable; no leaks when hot
PC-09	Nitrogen purge and blanket	Pressure-purge T-102 / V-101 and the amine storage with N2 to remove oxygen (oxidative degradation of MEA); keep a N2 blanket	SYS-04, 06	O2 below the target (calculator, part 5)
PC-10	Amine inventory make-up	Drain the water, charge MEA and demin water to 30 wt%; check with AIT-213 and lab titration	SYS-03…05	AIT-213 28–32 wt% (alarm limits); lab within ± 0.5 wt%

Amine inventory (PC-10)

Model hold-up (CO_2-free): T-101 sump 28.9 t + reboiler/T-102 sump 11.0 t + columns, exchangers, piping 50.0 t = **89.9 t** (MEA 26.98 t at 30.0 wt%)
First fill with 10 % margin (assumed): 98.9 t × 0.30 = 29.67 t MEA = **29.97 t of 99 % product** (assumed purity) + **68.9 t demin water**

Nitrogen pressure purge (PC-09)

$n = \ln(y_0/y_f)/\ln(P_H/P_L) = \ln(21/1)/\ln(200/105) = 4.72 \rightarrow$ **5 cycles** (P_H 200 kPa a below PSHH-301 250 kPa a; O_2 target 1 % assumed)
V = T-102 packed section 42.5 m³ + overhead gas volume 40 m³ (model) = 82 m³; $N_2 \approx 82 \times (200 - 105)/101.3 \times 5 =$ **387 Nm³**

6 · Commissioning steps mapped to the OTS procedures

Each commissioning step is practised on the simulator before it is done on the plant (Exercises → Procedures). Every step of the seven graded procedures is covered.

Step	Commissioning activity	SOP	OTS procedure	Procedure steps covered
C-01	Utilities in service, cooling water on, coolers in AUTO	SOP-01	Cold startup: circulate, heat, strip, then admit the flue gas	1. Cooling water on: CW supply running; TIC-202, TIC-302 and TIC-110 in AUTO
C-02	Solvent circulation established on amine: P-102 and P-101, FIC-201 ≥ 120 t/h, levels in AUTO	SOP-02	Cold startup: circulate, heat, strip, then admit the flue gas	2. Solvent circulating: P-102 and P-101 running, FIC-201 in AUTO ≥ 120 t/h, LIC-109 and LIC-208 in AUTO
C-03	Overhead system: P-103, LIC-303 and PIC-301 at 175 kPa a	SOP-02	Cold startup: circulate, heat, strip, then admit the flue gas	3. Overhead ready: P-103 running, LIC-303 in AUTO, PIC-301 in AUTO at 175 ± 5 kPa a
C-04	SIS reset and opened (permissives met), first live test of the damper and steam valve	SOP-01	Cold startup: circulate, heat, strip, then admit the flue gas	4. SIS reset and damper/steam opened (RUN)
C-05	First steam: heat the stripper on TIC-206 → FIC-207 cascade, strip the solvent to AI-203 ≤ 0.26	SOP-02	Cold startup: circulate, heat, strip, then admit the flue gas	5. Steam: TIC-206 in AUTO at 120 ± 1 °C with FIC-207 in cascade 6. Stripper boiling: TIC-206 within 1.5 °C of 120, PIC-301 175 ± 5, lean loading AI-203 ≤ 0.26
C-06	Water wash in service (P-104)	SOP-02	Cold startup: circulate, heat, strip, then admit the flue gas	7. Water wash in service: P-104 running
C-07	First flue gas: K-101 started, FIC-101 in AUTO, ramp to design rates	SOP-02	Cold startup: circulate, heat, strip, then admit the flue gas	8. Flue gas admitted: K-101 running, FIC-101 in AUTO ≥ 80 kNm³/h 9. Design rates: FIC-101 set point ≥ 170 kNm³/h and FIC-201 set point ≥ 260 t/h
C-08	Initial stable operation: hold at design and record the first data set	SOP-02	Cold startup: circulate, heat, strip, then admit the flue gas	10. Hold 10 min: capture ≥ 89 %, LIC-109 and LIC-208 30–70 %, AY-102 < 80 %, AY-212 ≤ 4.3 GJ/t
C-09	Pump trip and restart test (lean pump P-102); changeover to the spare when installed	SOP-08	Lean pump P-102 trip recovery	1. P-102 STOP and FIC-201 LO acknowledged 2. Steam cut back to stop over-stripping: FIC-207 in MAN ≤ 15 t/h 3. P-102 restarted 4. FIC-201 in AUTO ≥ 260 t/h 5. FIC-207 back in cascade with TIC-206 at 120 ± 0.5 °C 6. Hold 10 min: capture ≥ 88.5 %, LIC-109 and LIC-208 30–70 %
C-10	Load test: flue-gas CO2 4 → 6 % (part load with duct firing), lean flow and steam raised	SOP-06	Flue gas CO2 rises to 6 %: raise the solvent and the steam	1. Lean flow raised: FIC-201 set point ≥ 400 t/h 2. Reboiler set point 120–121 °C with FIC-207 in cascade (steam follows) 3. Hold 10 min: capture ≥ 85 %, AY-102 < 80 %, AY-210 < 85 %, TIC-206 ≤ 122 °C
C-11	Abnormal-situation drill in the absorber (diagnose, field remedy, stabilise)	SOP-09	Absorber upset, cause unknown: diagnose, remedy, stabilise	1. Upset diagnosed and the correct field remedy applied 2. Absorber hydraulics back to normal: AY-102 < 75 % and PDI-102 < 6 kPa 3. Hold 10 min: capture ≥ 89 %, AY-102 < 75 %, LIC-109 and LIC-208 30–70 %, AIT-213 29–31 wt%
C-12	Tuning for energy: SRD at ≥ 90 % capture (pre-test optimisation)	SOP-07	Optimisation: bring the SRD down at ≥ 90 % capture (live, 60 min)	1. AY-212 ≤ 4.00 GJ/t 2. Capture AY-112 ≥ 89.5 % 3. AY-102 and AY-210 below 75 % flood 4. No alarm active or unacknowledged
C-13	Planned shutdown test: gas out, strip lean, steam off, cool down with circulation; restart	SOP-03	Normal shutdown: gas out first, strip the solvent lean, then cool down	1. Flue gas reduced: FIC-101 set point ≤ 100 kNm³/h 2. K-101 stopped (flue gas to the stack) 3. Solvent stripped lean for the restart: AI-203 ≤ 0.23 with the steam still on 4. Steam off: FIC-207 in MAN at 0 % 5. Reboiler cooled below 100 °C with the solvent circulating 6. Solvent kept circulating (P-101 and P-102 running)

Step	Commissioning activity	SOP	OTS procedure	Procedure steps covered
C-14	Live ESD test from design rates (unit trip I-1), response and restart after the trip	SOP-04	Emergency shutdown	1. ESD within 60 s of the order: I-1 tripped 2. K-101 stopped and XV-207 steam confirmed shut 3. Solvent circulation kept running (P-101, P-102) 4. Reflux kept running (P-103) and PIC-301 in AUTO 5. All alarms acknowledged after the trip

7 · Performance test

Proves the design basis: **90.0 % capture**, **300 t/d** CO_2 product, steam **≤ 4.00 GJ/t**, MEA 28–32 wt%, held for **72 h** with the unit lined out first. Data: hourly averages; capture from AY-112 (AIT-101/111) cross-checked with AY-214 (FI-304). Acceptance values are assumed for teaching: capture and product from the design basis (K.cap, K.capTPD), SRD from the optimisation target, MEA from the AIT-213 alarm limits. A 72 h run is typical practice.

Criterion	Acceptance (assumed)	Model at design (noise off)
Capture AY-112	≥ 90.0 %	90.0 %
SRD AY-212	≤ 4.00 GJ/t	3.94 GJ/t
CO2 product FI-304	≥ 300 t/d	300 t/d
MEA AIT-213	28–32 wt%	30.01 wt%
Duration	72 h continuous	—

8 · Handover

Certificate / milestone	Condition	When
Mechanical completion certificate (per system)	All punch A cleared; test packs signed	Before pre-commissioning of that system
Ready for start-up (RFSU)	Pre-commissioning PC-01…PC-10 complete; SIS proof tested; operators trained (OTS records)	Before amine and steam are introduced
Performance test certificate	Part 7 criteria met over the test duration	After C-12
Provisional acceptance (handover to operations)	Punch B list agreed, as-built documents, spare parts, vendor manuals, training records	After the performance test
Final acceptance	Punch B closed, warranty period complete	End of warranty (typical 12–24 months, assumed)

9 · Engineering queries

No.	Query	Documents	Suggested resolution
EQ-CM-01	PSVs on T-102, V-101, E-102 shell and E-101 are on HOLD; a system cannot be leak-tested above its relief set or reach RFSU without its relief devices.	CC-PID-101 note 5; this plan PC-04; Relief & vent	Close the relief load basis and install the PSVs before PC-04 / RFSU of SYS-04 and SYS-06 (punch A).
EQ-CM-02	No nitrogen connection is drawn, yet purging and blanketing (PC-09) need one on T-102 / V-101 and the amine storage.	CC-PID-101 (no N2) vs this plan	Add N2 utility connections at rev B.
EQ-CM-03	Amine storage, make-up pump and filter are not shown (note 7) but the first fill (PC-10) depends on them.	CC-PID-101 note 7 vs PC-10	Issue the solvent-management sheet before procurement of PKG-09.
EQ-CM-04	Performance-test acceptance criteria (capture, SRD, product, duration) are not in any issued document; this plan assumes them.	This plan part 7 vs the design basis	Agree the guarantee values and test procedure in the contract.
EQ-CM-05	The materials (note 6) are TBC, so the passivation scope (PC-03) is provisional.	CC-PID-101 note 6 vs PC-03	Freeze the materials selection report.
EQ-CM-06	No spare pumps on the P&ID;: the pump changeover in SOP-08 and commissioning step C-09 cannot be done as written.	CC-PID-101 vs SOP register vs load list	Decide the sparing philosophy (EQ-EE-02).